

AI-BFSI Internship Project Documentation

Performance Testing Report – Subscription Waste Detector

1. Introduction

Performance testing evaluates how efficiently an application performs under expected operating conditions. It measures responsiveness, stability, and resource utilization while executing its core functionalities.

For the **Subscription Waste Detector**, testing focused on verifying responsiveness during normal usage: file upload, data processing, dashboard rendering, visualization generation, AI recommendation response, and overall stability. Tests were conducted using realistic sample datasets representing everyday subscription expense records.

2. Objectives of Performance Testing

- Evaluate application responsiveness
- Verify file upload performance
- Measure data processing efficiency
- Validate dashboard rendering
- Assess visualization generation
- Evaluate AI recommendation response
- Confirm stable operation under normal usage

3. Testing Environment

Component	Configuration
Application	Subscription Waste Detector
Platform	Streamlit Web Application
Programming Language	Python
Data Processing	Pandas
Visualization	Plotly
AI Service	Groq API
Browser	Google Chrome
Operating System	Windows
Dataset	Sample CSV and Excel expense records

4. Performance Test Scenarios

Test ID	Test Scenario	Expected Result
PT-01	Launch application	Application loads successfully
PT-02	Upload CSV file	File imported successfully
PT-03	Upload Excel file	Spreadsheet processed correctly

PT-04	Manual expense entry	Record added successfully
PT-05	Subscription detection	Recurring subscriptions identified
PT-06	Dashboard generation	Dashboard displayed correctly
PT-07	Plotly visualization	Charts rendered successfully
PT-08	AI recommendation generation	Recommendations displayed successfully

5. Performance Test Results

Test Case	Result	Status
Application startup	Successfully launched	Pass
CSV upload	Imported without errors	Pass
Excel upload	Imported without errors	Pass
Manual expense entry	Processed successfully	Pass
Expense analysis	Completed successfully	Pass
Subscription detection	Correctly identified recurring subscriptions	Pass
Dashboard rendering	Displayed correctly	Pass
Plotly charts	Rendered successfully	Pass
AI recommendations	Generated successfully via Groq API	Pass

6. Functional Performance Evaluation

- **Application Startup:** Streamlit initialized successfully and displayed the UI without errors.
- **File Upload Performance:** CSV and Excel files uploaded smoothly; supported formats validated.
- **Data Processing:** Pandas processed records efficiently; no errors with sample datasets.
- **Subscription Detection:** Recurring payments identified correctly.
- **Dashboard Rendering:** Summaries, reminders, and recommendations displayed as expected.
- **Visualization Performance:** Plotly charts rendered interactively and responsively.
- **AI Recommendation Performance:** Groq API returned suggestions successfully; response time dependent on internet connectivity.

7. Resource Utilization

- Stable memory usage with sample datasets
- Smooth dashboard interaction
- No interface freezing observed
- Successful file processing
- Consistent visualization rendering

8. Performance Limitations

- Performance depends on dataset size
- AI recommendation response depends on Groq API availability
- Internet connectivity required for AI features

- Larger datasets may increase processing time
- Enterprise-scale workloads not evaluated (outside scope)

9. Performance Summary

Performance Area	Observation
Application Startup	Stable
File Upload	Successful
Data Processing	Efficient for project datasets
Subscription Detection	Accurate
Dashboard Rendering	Responsive
Plotly Visualization	Smooth
AI Recommendation	Successful (Internet dependent)
Overall Stability	Stable during testing

10. Recommendations

Future improvements could include:

- Optimizing for larger datasets
- Asynchronous AI requests for smoother UX
- Performance logging for key stages
- Caching mechanisms for repeated analysis
- Enhanced handling of very large transaction files

11. Conclusion

Performance testing confirmed that the **Subscription Waste Detector** operates reliably under normal usage conditions. It successfully processes uploaded records, detects subscriptions, generates dashboards, and produces AI-powered recommendations without functional issues.